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Comparative study of deep learning techniques for DeepFake video detection

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Abstract

Deep learning addresses a wide range of complex challenges, spanning from computer vision to data analytics. It is also employed to develop softwares that pose threats to privacy and security. To develop a DeepFake video, an individual in the original video is replaced with someone else using deep learning. Various deep learning-based techniques have been proposed to detect DeepFakes. In this work, we extensively analyse DeepFake video detection techniques considering their strengths and limitations. We provide a comparative analysis along with discussing their architectures and performances. Finally, we propose hyperparameter settings that improve deep learning model's overall accuracy and efficiency.

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Keywords: DeepFake; Artificial intelligence; Deep learning; Convolutional neural networks; Generative adversarial networks; Recurrent convolutional neural networks

1. Introduction

Synthetic media, advanced through the convergence of machine learning and artificial intelligence technology, has enabled advancements in various fields, i.e., interactive learning, media production, content creation, marketing, and healthcare among many others. It involves the artificial production, manipulation, and modification of data and media (i.e. audio, video, and images) to facilitate image synthesis, audio synthesis, speech synthesis, video synthesis, and natural language generation [1]. On the other hand this has led to many ethical questions related to the privacy and authenticity of the digital contents. Sythetic media allows a user to digitally manipulating and overlaying the appearance of one person onto another's body, also known as a DeepFake.

The word "DeepFake" was coined by an unknown person on Reddit in 2017 [2]. The creation of DeepFake videos [3] include training a Deep Learning (DL) algorithm on datasets of the target person's videos. The algorithm then replicates the target individual's facial expressions and movements to integrate them into a fake video. Due to the convincing impersonation, it threatens the privacy and security of individuals. Additionally, cyber-criminals orchestrate phishing scams or identity theft operations with alarming precision using DeepFakes. People purposefully spread false information or they may have malicious intent with deepfakes. The content is designed to harass, intimidate, demean, and undermine individuals. The fake videos not only generate misinformation and confusion about crucial matters but also stimulate other unethical behaviours such as creating revenge porn or identity theft.

DeepFake algorithms requires a large amount of video and image data to creating photo-realistic videos and images for model training. The underlying enabling technology in DL is neural network that consists of interconnected nodes in a layered structure. Two types of neural networks are used to create DeepFake remarkably lifelike videos: one generates the fake content, while the other evaluates its realism. These two networks work together in a method termed Generative Adversarial Networks (GANs).

The growing complexity of DeepFake creation makes the identification task more difficult. The pervasive availability of user-friendly deepfake technology, which greatly aids in the

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Table 1
Comparison of our study with the literature.

Article and year	DeepFake detection methods	Comparison of DL algorithms	Limitations of current methods	Future directions
[4]-2020	✓	✓	✓	×
[5]-2021	✓	✓	×	×
[6]-2022	✓	✓	✓	×
[7]-2022	✓	✓	×	×
[8]-2022	✓	×	×	×
[9]-2022	✓	✓	×	×
[10]-2023	✓	✓	×	×
This work-2024	✓	✓	✓	✓

training of DL models, to those with low technical proficiency simplifies the creation of persuasive deepfake videos. In addition, the vast online content poses a significant challenge in manually confirmation the genuineness of each video, therefore requiring the creation of automated detection techniques. Given this difficulty, it underscores the importance of studying methods to identify them accurately [11,12].

DL is used not only for creation but also for the detection of DeepFakes. This entails training DL models using extensive datasets including both authentic and manipulated images, audios or videos. These methods detect tiny anomalies or discrepancies in the produced material that may not exist in the authentic videos. The anomalies could be in the face expressions, abnormal motions, or irregularities in illumination and shading. Additionally, many techniques prioritise the examination of certain attributes such as frequency of blinking, characteristics of the skin, or motions of the lips. Facial recognition is used to determine if the faces in the video match those of recognised persons, aiding in the identification of DeepFake videos. In this comparative study, we discuss the major DL techniques used to detect the DeepFake videos. Table 1 highlights and compares the salient features of our survey with the previous literature.

In this work, our main contributions are, (1) We review the background of DL for DeepFake detection highlighting prominent DeepFake detection methods. (2) A comparative analysis of DL algorithms for DeepFake detection is provided that highlights the constraints of existing methods, deliberate on potential solutions, and delve into prospective research directions. (3) Given the accelerating sophistication of DeepFake technology, our survey seeks to distill actionable insights into video-based detection methods.

The rest of the paper is organised as follows: Section 2 presents a meticulous review of the existing literature, highlighting key studies in the realm of DeepFake generation and detection methods. Section 3 delineates the types of DeepFake and Section 4 discusses DeepFake creation techniques. In Section 5, we delve into a thorough discussion on the methods for DeepFake detection. It also discusses the role of DL for DeepFake detection. Section 6 concludes the paper and highlights the details of the existing methods.

2. Background

Identifying Deepfakes presents a significant challenge still its detection is essential for effective digital investigations. Ali et al. [13] present a study that aims to detect DeepFake using a new method called DeepFake Predictor (DFP), which combines VGG16 and CNN architecture. They use a dataset with real and fake faces and compare their approach with other techniques like Xception, NAS-Net, Mobile Net, and VGG16. DFP method achieves 95% precision and 94% accuracy. In [14], authors create a voting system that combines multiples methods for fake video detection. They use datasets from FaceForensics++, which had videos made by four different DeepFake technologies. To tackle DeepFake, authors [15] introduce a technique that combines different DL models to spot manipulated videos. Proposed method shows better results than comparative methods, reaching 99.65% accuracy and a perfect AUC score in detecting DeepFake. Amritpal et al. [16] develop a model that pays attention to smaller differences in the frames. They test the model on a huge dataset called deep fake detection challenge and find their method reaching 97.6% accuracy in spotting the fake videos. David et al. [17] suggest a way to automatically find these fake videos using a process that understands time in videos. They use CNN to understand each frame and then train another neural network called RNN to explore if a video is manipulated or not. Previous works employ various types of databases for DeepFake studies. Examples of some databases used in previous DeepFake studies are provided in Table 2.

Authors [18] propose a method that introduces a new neural network-based method to find fake videos. They use a technique to pick out important frames in videos, reducing the work needed to spot DeepFake. Authors [19] introduce a method using facial recognition and learning from facial expressions and movements to detect face-swapped DeepFake. This technique combines recognising faces with understanding behaviour using a CNN. Zhao et al. [20] propose an Interpretable Spatial-Temporal Video Transformer that utilises a decomposed spatial–temporal self-attention and a self-subtract mechanism. This approach captures spatial artifacts and temporal inconsistencies for robust detection.

3. Types of DeepFake

DeepFakes are divided into different categories including photo DeepFakes, audio DeepFakes, video DeepFakes, and audio and video DeepFakes as in Fig. 1 . We further elaborate these categories of DeepFakes in the following subsections.

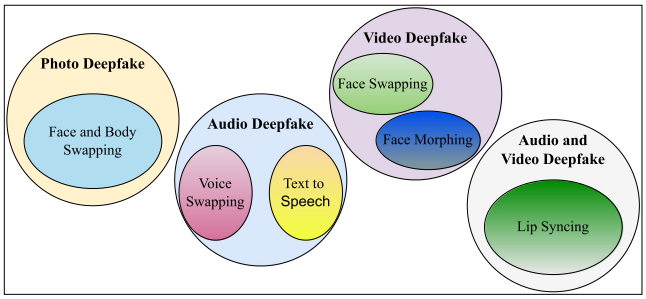


Fig. 1. Types of DeepFake.

3.1. Photo DeepFake

Face and Body Swapping: In this technique, alterations are made to a person's face and body by substituting or combining them with features from another person. This process creates an entirely new individual within the initial image. DeepNotch [21] presents ongoing competition between creating highly realistic DeepFake images and developing accurate detection methods. DeepNotch introduces a simple yet effective pipeline that reduces artifact patterns in fake images without compromising quality. By employing spatial-domain notch filtering through a combination of overwhelming spatial noise and deep image filtering, DeepNotch significantly decreases the accuracy of three fake image detection methods.

3.2. Audio DeepFake

Voice Swapping: In this method, the original voice in an audio is switched with someone else's voice through imitation. This technique has potential for enhancing audiobooks by imitating various voices, both male and female, in different accents. This provides listeners with a richer experience while enjoying the book.

Text to Speech: In this process, AI converts written text into spoken audio. It transforms text into various voices and accents. For instance, there was an incident where someone created a contentious recording using the voice of Jordan B. Peterson, a well-known professor. This technique is handy for fixing errors in spoken dialogue within a movie script without needing to re-record the lines.

3.3. Video DeepFake

The DeepFake videos are created in two major ways (i) Face swapping and (ii) Face morphing. Both of these techniques operate in different ways. The main difference lies in the way these techniques manipulate and transform facial features.

Face Swapping: Face swapping algorithms are used to substitute the face of one individual in a video with the face of another individual, all while preserving the original person's facial emotions, head movements, and other facial dynamics. The program utilises a DL model to acquire knowledge about the facial characteristics and emotions of both persons. The model exchanges the facial appearances by transferring the distinctive characteristics of one individual's face to another person in the fabricated video. This is illustrated in the DeepFake creation section where we show how Amy Adams' face is replaced with Nicolas Cage's face in Fig. 14.

Face Morphing: Face morphing is the process of seamlessly combining the face characteristics of two or more persons to achieve a gradual and fluid transformation. The algorithm identifies and isolates important face characteristics from the original photos of the specified people. In the fake video, the algorithm progressively morphs one face into another by seamlessly interpolating between the facial features via the use of methods such as picture warping and blending. Forensic Symmetry (FS) [22] introduces a novel DeepFake forensics method to assess whether symmetrical face patches contain the same or different natural features. The approach utilises a multi-stream learning structure with two feature extractors—one for symmetry features from front face images and another for similarity features from side face images. The system maps symmetrical face patch pairs into an angular hyperspace to quantify differences in their natural features.

Table 2
Databases used in DeepFake studies.

Database name	Source
DeepFaceLab Dataset	Online repositories
CelebA Dataset	CelebA Dataset Website
FaceForensics++	TU Munich
DFDC Dataset	Kaggle
UADFV	Utrecht University

3.4. Audio and video DeepFake

Lip Syncing: In this method, not just the face but also the mouth movements of one person are synced to match another person's speech. The words spoken by the creator are voiced in the victim's voice. An example of this is seen in the video "You Won't Believe What Obama Says in This Video!" made by Jordan Peele. This technique is handy for translating instructional and advertisement videos into various languages while maintaining the same voice and original video.

4. DeepFake creation

The creation of DeepFake videos involves DL, which is really good at handling complex information. One specific kind of DL, called deep auto-encoders, is often used to make these fake videos. DeepFake uses a special pairing of auto-encoder and decoder to create these fake videos. Two auto-encoders take important information from face pictures, and then another part of the system rebuilds the pictures in that style. To switch faces between two pictures, two sets of these systems are required—one for each picture. They learn from a bunch of pictures, and they share some important settings between the two sets. During training, two networks use one encoder but different decoders, which can be seen clearly from Fig. 2.

Table 3
Commonly used DeepFake creation tools.

Tool	Reference	Characteristics
DeepFaceLab	[23]	Multiple methods of face extraction are supported.
Faceswap-GAN	[24]	Utilises auto-encoder architecture with adversarial and perceptual loss.
Faceswap	[24]	Uses two pairs of encoder–decoder for face swapping.
Few-Shot Face	[25]	Employs latent embeddings extraction for GAN processing using a pre-trained face recognition model.
DFaker	[26]	Implements a reconstruction of faces using a loss function called DSSIM.
DeepFaketf	[26]	Similar to DFaker but utilises Tensorflow structure.
AvatarMe	[27]	Creates 3D faces from arbitrary "wild" images.
StyleRig	[28]	Allows self-monitoring without the need for annotations.
MarioNETte	[29]	Adapts identity without requiring further fine-tuning.
DiscoFaceGAN	[30]	Incorporates 3D priors in adversarial processes.
StyleGAN	[31]	Automatically separates high-level properties in a new architecture.
Face2Face	[32]	Enables real-time reenactment of a monocular target video.
NeuralTextures	[28]	Learns feature maps during scene capture and stores them on 3D mesh proxies.
Transformable Bottleneck Networks	[33]	Allows fine-grained 3D image modification.
Neural voice puppetry	[34]	Synthesises audio-driven facial videos.

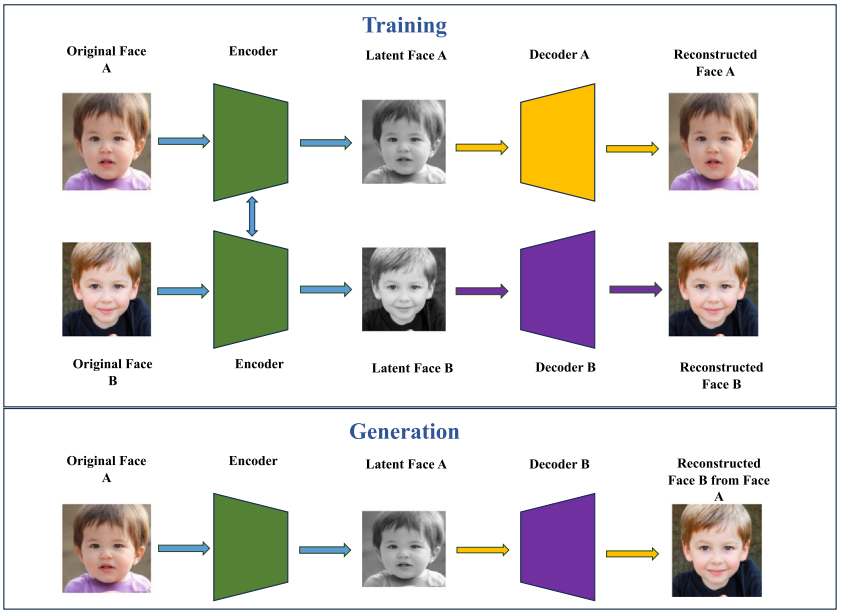


Fig. 2. Two encoder–decoder pairs are used in this DeepFake production strategy.

In simpler terms, both sets of encoder networks are exactly the same. This helps the shared encoder figure out and remember how similar two sets of face images are. Previous techniques, such as Deep-FaceLab, DFaker, and DeepFake tf, use this method. An improved version of this technique, faceswap-GAN, adds extra ways to make the fakes better by using adversarial and perceptual losses from VGGFace along with the encoder–decoder system.

A framework consisting of an auto-encoder and a decoder is developed to aid in the process of creating fake pictures. An auto-encoder extracts latent characteristics of the face such as facial expression, skin tone, eye colour, and face angle, and then the decoder reconstructs the picture. Due to the encoder's difficulty in remembering all the characteristics, it gets just those that are necessary for reconstructing the actual picture. When the faces are swapped, two encoder and decoder pairs are utilised, with the encoder's features being shared across two network pairs. The common encoder's role in this process is to seek for and train itself on the comparable characteristics of both faces, which is not difficult given that all faces have common features such as nose, mouth location, and eyes.

Rossler et al. [35] describe the process of creating a fake face may be classified into two categories: facial expression manipulation and facial identity modification. In the first, it changes the facial expression of one face to that of another. The most often used method is Face2Face (F2F), which simulates the facial expressions of a monocular target video stream in real time. There are various tools used for the creation of DeepFakes. A summary of the tools and their salient features is provided in Table 3.

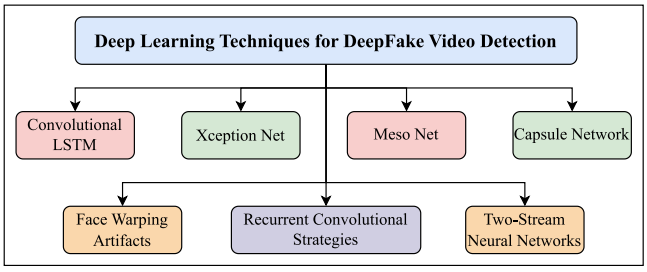


Fig. 3. Common deep learning techniques used for DeepFake video detection.

5. DL for DeepFake detection

DL techniques are involved in both creating and detecting DeepFakes, as they pinpoint particular and meaningful features within fake videos, distinguishing them from both the original and modified versions. The process of detecting DeepFake videos using DL consists of two main categories: (i) visual artifacts within the frame and (ii) temporal features beyond the frame. First category refers to the detection of anomalies present within the video frame itself, such as unnatural facial movements or discrepancies in lighting and shadows. Deep classifiers are employed to analyse these visual artifacts and identify signs of manipulation. The second category involves the detection of aspects of the video that extend beyond individual frames, such as the continuity of motion and coherence of actions over time. Recurrent classification models are utilised to analyse temporal patterns and detect any irregularities that may indicate the presence of a DeepFake. To lessen the impact, several large corporations have developed their own DeepFake-filtering techniques. Fig. 3 displays the major DL based DeepFake detection techniques provided in our comparative study.

DL and big data advancements have made it easier to spread fake videos online. In [36], Gambin et al. discuss how Deepfakes are made, and efforts to detect them. Researchers also examine how Deepfakes could affect social media and connected devices. Heidari et al. [37] discuss about the deepfakes which are created by some digital tech developments can be risky. This study discusses how we detect deepfakes using DL. They sort these methods into categories like video detection and image detection. They also give readers a better understanding of how deepfakes are made and found, the latest progress in this area, what is not working well in current security methods, and what areas need more study. Most of the research uses CNN for deepfake detection, especially for videos.

Authors in [38] explore the fast-changing world of deepfake technology. Authors look at both making and spotting deepfakes. Authors discuss about advanced methods like Cramer-GAN and StyleGAN for making deepfakes. Each method has its strengths, like StyleGAN's precise control over faces and PGGAN's high-quality images. Authors also discuss about the tools like ProGAN and CLRNet for finding deepfakes. Researchers proposed new methods for making and spotting deepfakes, giving a complete picture of this rapidly changing field. Space-Frequency Interactive Convolution (SFIConv) [39] addresses the concerns surrounding DeepFake and proposes a novel approach to enhance DeepFake detection. Unlike existing methods that improve traditional backbone networks, SFIConv constructs a new backbone network with Multichannel Constrained Separable Convolution to efficiently capture manipulation clues in the space-frequency domain. We now discuss some of the most popular and cutting-edge DL based DeepFake detection techniques.

5.1. Convolutional LSTM

In this technique [17], a temporal-aware pipeline is used to identify DeepFake movies. A convolutional LSTM structure is employed to process frame sequences in the proposed scheme. In a convolutional LSTM, there are two main components, (1) Frame feature extraction using CNN, (2) For temporal sequence analysis, using LSTM.

Given an image of a series, a convolutional LSTM is utilised to build a temporal sequence descriptor for image processing of the shot frame. The shallow network comprises two linked layers and one dropout layer to eliminate overfitting during the training process. The convolutional LSTM is split into two parts: a CNN and an LSTM. Fig. 4 displays the architecture of CNN and LSTM.

InceptionV3 [40] is used, but without the fully-connected placed on top, which leverages the ImageNet pre-trained model to construct a deep representation for every frame. The 2048-dimensional feature vector is utilised as the sequential LSTM input after the final pooling layer.

5.1.1. LSTM for sequence processing

A 2048-by-2048 LSTM with a 0.5 dropout probability is employed, which recursively analyses sequences in a meaningful way. The LSTM model is given a succession of 2048-dimensional ImageNet feature vectors. The LSTM is followed by a 512 fully connected layer with a 0.5 chance of dropout. Finally, the probability of the frame series being pristine or DeepFake are computed using a SoftMax layer.

5.1.2. Dataset and results

300 DeepFake movies are acquired from several video-hosting services for this study, as well as 300 more videos from the HOHA [41] dataset, resulting in a total of videos in the set of data to 600.

Because this contains a genuine collection of sequence clips from well-known movies with a focus on human behaviour, the HOHA dataset is chosen as a source of clean videos. The dataset is randomly split into 70/15/15 sets, used for training, validation, and testing. For balanced splitting, 300 DeepFake videos are split first and the process is repeated for 300 non-manipulated videos, making a final set containing exactly 50% videos of each class. In terms of video sequence data preprocessing, following settings are followed: (i) Taking the mean of each channel and subtracting it. (ii) Every frame has been resized to 299x299 pixels. (iii) The length of the input

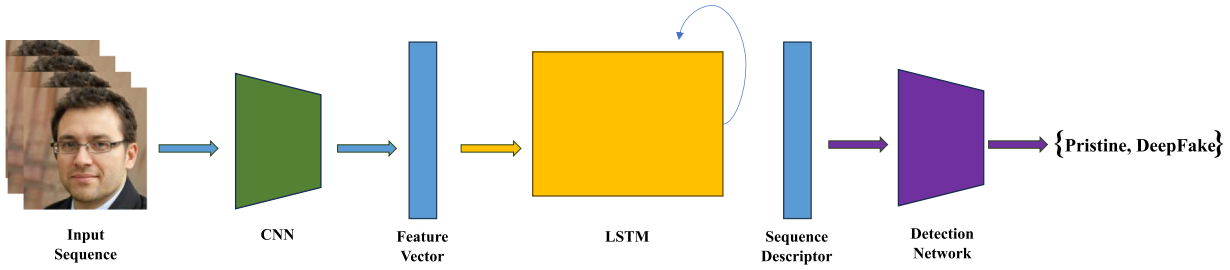


Fig. 4. CNN LSTM architecture.

Table 4
Convolutional LSTM results.

Method	Train acc.(%)	Val acc. (%)	Test acc. (%)
Conv-LSTM, 20 frames	99.5	96.9	96.7
Conv-LSTM, 40 frames	99.3	97.1	97.1
Conv-LSTM, 80 frames	99.7	97.2	97.1

sequence is controlled by sub-sequence sampling of length -N, N = 20, 40, 80 frames. (iv) The optimiser is set to Adam [42].

The efficiency of convolutional LSTM method in terms of detecting accuracy is presented in the Table 4. The convolutional LSTM differentiates whether the fragment being analysed originates from a fake video or not with only 2 s of footage, as shown in the data (40 frames for videos sampled at 24 frames per second).

5.2. XceptionNet

XceptionNet [43] is a complex CNN structure specifically created for the purpose of image classification tasks in order to detect manipulated faces. It is renowned for its capacity to acquire hierarchical representations of characteristics in images, making it well-suited for tasks such as facial recognition and detection of image modification. It also presents FaceForensics++ dataset while discussing manipulation creation and forgery detection techniques. FaceForensics++ is a supervised dataset designed specifically for training and evaluating deep-learning algorithms to detect and classify manipulated facial images in fake videos. The dataset comprises instances of altered facial images generated by many methods, such as Face2Face, FaceSwap, and NeuralTextures. Each of Swap strategies signifies a distinct method of altering face content.

5.2.1. Face2Face

(F2F) [44] is a face regeneration method of transferring the expressions of a source video to a target video while maintaining the identity of the target individual. It can also be described as a graphical approach for moving the facial region of a video from one to another. From the original video data, it constructs a neural texture of the target individual, including a rendering network.

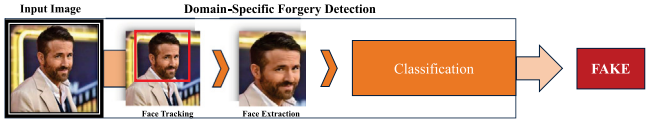


Fig. 5. Domain-specific forgery detection.

5.2.2. FaceSwap

The Faceswap [45] is a technique based on DL to detect manipulated faces. CNN architecture plays a crucial role in this process by analysing image features to distinguish between authentic and manipulated facial content. FaceForensics++ also provides a dataset of manipulated faces for Faceswap. The combination of XceptionNet and FaceForensics++ enables the detection of Faceswap manipulations within fake videos.

5.2.3. NeuralTextures

NeuralTextures [28] is a technology that use DL techniques to create lifelike textures and intricate details, such as facial characteristics. The XceptionNet model is used inside a detection system to accurately identify manipulations, namely those that are created using NeuralTextures. NeuralTextures modifications include generating artificial face information that is so lifelike that it becomes difficult to differentiate from genuine video footage. The FaceForensics++ dataset contains instances of alterations generated using NeuralTextures. These samples are used as training data to construct algorithms that can recognise alterations made by NeuralTextures.

Apart from the above mentioned techniques, there are some other models which are used to detect forgery in DeepFake videos. One of them is domain specific forgery detection model. In this model, a face tracking technique is used to extract domain specific information from input sequences, detect a face in a video, and extract the face region. Conservative crop is applied, and the face is reconstructed. The face region is passed to classification network which outputs the prediction. Fig. 5 illustrates the working principle of domain-specific forgery detection model. Using the described workflow, various variations of state-of-the-art classification techniques are assessed on the dataset.

5.2.4. Dataset and results

All the variants of the approach are trained on their created benchmark: large-scale manipulation database, which include DeepFake (DF), F2F (F2F), FaceSwap (FS) and NeuralTextures (NT). The results of the methods are given in the Table 5.

Table 5
Results of different models trained on FaceForensics++ dataset.

Method	DF	F2F	FS	NT
Cozzolino et al.	85.45	67.88	73.79	78.00
Rahmouni et al.	85.45	64.23	56.31	60.07
Bayar and Stamm	84.55	73.72	82.52	70.67
MesoNet	87.27	56.20	61.17	40.67
XceptionNet	**96.36**	**86.86**	**90.29**	**80.67**

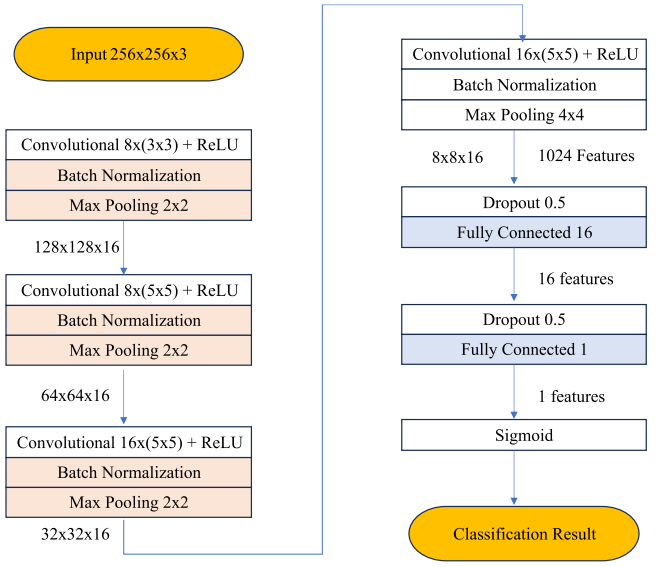


Fig. 6. The network architecture of Meso-4 [46].

5.3. MesoNet

MesoNet [46] presents an efficient method to detect face tampering in videos. DL technique is used to detect facial video forgery using two networks, each of which have a small number of layers and are based on the mesoscopic properties of photographs. The two following architectures receive the highest classification scores using a minimal amount of representation and a limited number of parameters. Meso-4 and MesoInception-4 are advanced DL models created as part of the MesoNet project.

5.3.1. Meso-4

This network starts with four successive convolutions and pooling layers, then moves on to a dense network of one hidden layer. To enhance generalisation, convolutional layers incorporate ReLU activation functions with non-linearities, and to prevent vanishing gradient effect, Batch normalisation is used. Dropout of 0.5 probability is used by FC layers to regularise and enhance robustness of the network. Total number of trainable parameters for this network are 27,977. Fig. 6 illustrates the network architecture of Meso-4 and Fig. 7 depicts the architecture of MesoInception-4.

5.3.2. MesoInception-4

There is an alternate approach for replacing the Meso-4 inception module's first two convolutional layers. Objective

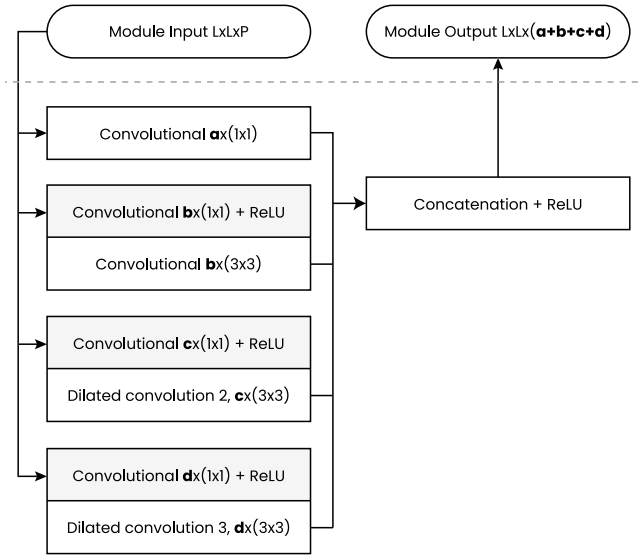


Fig. 7. MesoInception-4 architecture [46].

of this inception module is to increase the feature space by stacking the contribution of multiple convolutional layers with different kernel shapes. Instead of 5×5 convolutions, 3×3 dilated convolutions are employed to prevent high semantic. 1×1 convolutions are added before dilated convolutions for dimensionality reduction and 1×1 convolution is added in parallel which serves as a skip connection between modules. The classification results are not improved by replacing more than two layers with inception modules. The parameters chosen for this module at layer i are (a_i, b_i, c_i, d_i). The total number of trainable parameters for this network are 28,615.

5.3.3. Dataset and results

175 forged video rushes are gathered from various sites. Their length varies between two and three minutes, with a minimum resolution of 854 480 pixels. The H.264 codec is used to compress all the videos using various compression levels. The Viola–Jones detector is used to extract all the faces, which are then aligned using a trained neural network for detecting facial landmarks. The dataset is then doubled by include genuine face pictures, which are likewise collected from various websites and has the same resolutions as the originals. Finally, it is checked for misalignment and improper facial detection by hand. Face-Forensics dataset [35] is also used, which contains more than thousand fake videos and their originals using F2F approach. Out of over a thousand videos, only 300 are used for training. The 150 fake videos and their original are used for model evaluation.

With consecutive batches of 75 pictures of size $256 \times 256 \times 3$, the network's weights are optimised using the ADAM optimiser with default settings (β_1 = 0:9 and β_2 = 0:999) The original learning rate of 10^{-3} is divided by 10 each 1000 iterations, resulting in a learning rate of 10^{-6}. To improve generalisation and robustness, input batches are exposed to a variety of small random changes, such as zooming, rotating,

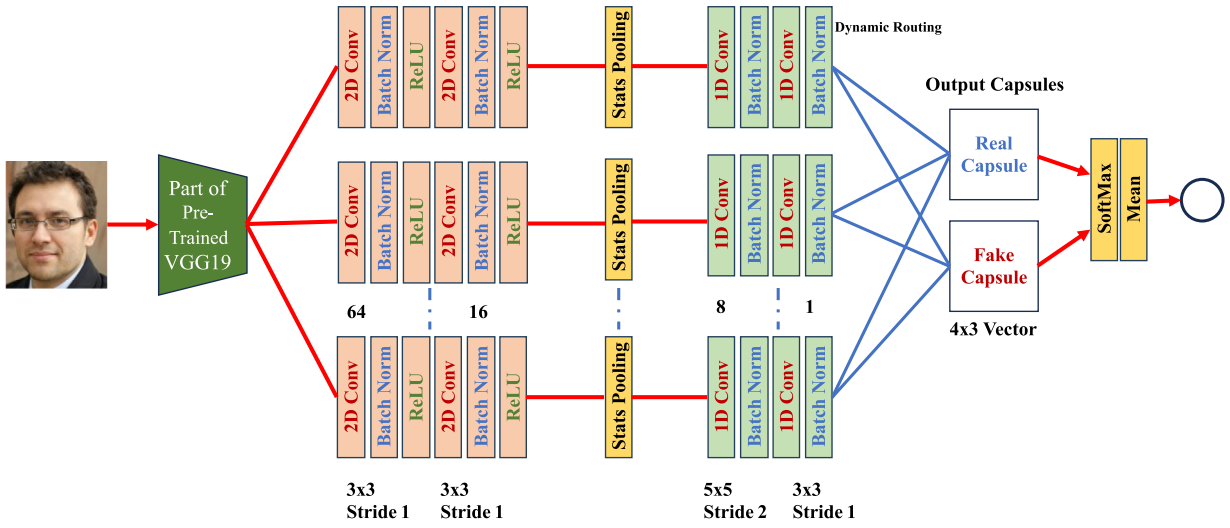


Fig. 8. Capsule-Forensics architecture.

Table 6
Aggregation score of the inception modules.

Model	Accuracy on DF dataset (%)	Accuracy on F2F dataset (%)
Meso-4	0.969	0.953
MesoInception-4	0.984	0.953

horizontal flipping, and altering brightness and colour. As shown in Table 6, the techniques have a detection rate of 98 percent for DeepFake movies and 95 percent for F2F videos in real-world internet dispersion situations.

5.4. Capsule network

Capsule network [47] identifies a wide range of assaults, from presentation assaults including printed graphics and re-played videos to DL-based attacks involving false videos. A lightweight, detector is developed that identifies any type of assault and is as effective as task-specific detectors. Mostly CNNs with a wide range of parameters are used in most cutting-edge detectors, but Capsule network uses a new type of CNN with imposing results in the domain of computer vision. Capsule forensics DeepFake detection architecture is shown in Fig. 8.

Input data is pre-processed, frames are separated if the input data is a video. If the aim is to recognise computer created graphics, frames are separated into patches (or frames). In order to discover a false face or faces, the face area is clipped using a face detection algorithm. The picture size selected is 300 by 300, since this figure is an even number and is large enough to provide enough information for detecting phoney materials. The picture that is about to enter the capsule network goes via a portion of the VGG-19 network that is trained on the ILSVRC.

From the first to the third max pooling layer, the VGG-19 network is deployed. Many main capsules and two output capsules make up the capsule network ("real" and "fake"). Three capsules are employed in light networks (which need less memory and calculation), whereas 10 capsules are used in the heavy networks (It needs more memory and compute but yields better results). Each main capsule consists of a 2-dimensional convolutional layer, a statistical pooling layer, and a 1-dimensional convolutional layer. In forensics, the statistical pooling layer is believed to be quite effective. The post-processing unit is the last component. The scores of the gathered patches are averaged if the goal is to detect entirely computer-generated pictures. When using video as an input, the scores of all frames are averaged.

Initially, the capsule-forensics network consists of three primary capsules for detecting fake images. Dropout and random noise are added to the network during training to reduce overfitting. The images are resized to 300 × 300. Second network is trained for detecting fake videos and the network consisted of ten capsules with larger input size (300 × 300) and dropout applied in training.

5.4.1. Dataset and results

The network is evaluated on DF, F2F and FaceSwap dataset as shown in Table 7. To make the results more convincing, the network is evaluated on a large and more challenging dataset FaceForensics++ database. Twenty-five epochs are used to train all Capsule-Forensics versions.

5.5. Face warping artifacts

Face Warping [48] presents a DL-based method distinguishing fake videos from the original ones as shown in Fig. 9. The technique is based on a feature of DeepFake videos: due to computational constraints and processing time, the DeepFake algorithms synthesise only a fixed size of face images, which must be affine warped to fit the source's face configuration.

Due to the variance in resolution, warping causes distinct artifacts, which are then used to identify DeepFake. A specialised CNN model is used to identify such artefacts by comparing the produced face regions with their surrounding

Table 7
Accuracy of Capsule Forensics on FaceForensics++ database.

Model	Real data (%)	DF (%)	F2F (%)	FaceSwap (%)
Capsule-Forensics + Dropout + Noise (300 x 300)	89.57	92.17	90.00	92.79
Capsule-Forensics + Dropout + Noise (video)	89.57	92.17	90.36	92.79

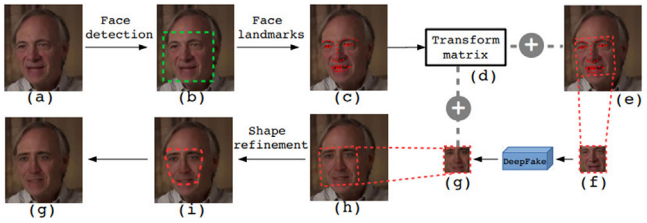


Fig. 9. Face warping pipeline.

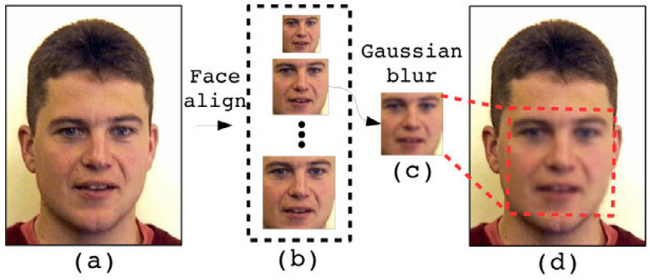


Fig. 10. Overview of negative data generation.

area. The CNN model is trained more quickly by modelling the resolution inconsistencies in affine face warpings directly. First, the face detection is performed, and then the location of landmarks is determined to calculate matching transform matrices for the faces. The faces that are aligned are blurred before they are deformed using the affine transformation matrix to return to the original picture. To mimic various resolution situations of distorted faces, faces are aligned into varied sizes to improve data variety. Negative data is created using simple image processing techniques, saving time and compute resources as shown in Fig. 10.

5.5.1. Dataset and results

As positive examples, the CNN model is trained using 24,442 JPEG face images gathered from the internet. DeepFake algorithms are used to generate negative examples, but that require computational resources. So, the negative examples in this study are generated by directly simulating affine face warping. Four CNN models including VGG-16, ResNet50, ResNet101 and ResNet152 are trained on the training dataset. Table 8 discusses the results regarding AUC performance of the face warping artifacts models.

Table 8
AUC performance of the models.

Models	UADFV (%)	DeepFakeTIMIT LQ (%)	DeepFakeTIMIT HQ (%)
VGG-16	84.5	84.6	57.4
ResNet-50	97.4	99.9	93.2
ResNet-101	95.4	97.6	86.9
ResNet-152	93.8	99.4	91.2

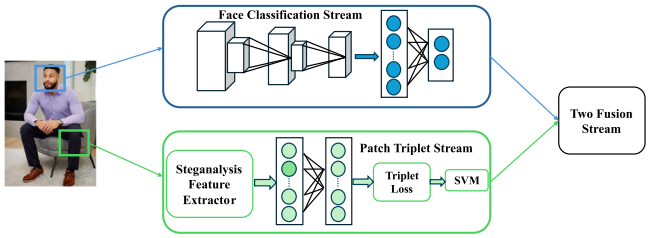


Fig. 11. Overview of Two-Stream Network.

Faces in the pictures are identified, and face areas are extracted. Faces are arranged in various sizes, with one scale selected at random. Gaussian blur with a 5 × 5 kernel size is applied to smooth the image. In affine warped faces, this approach produces additional resolution situations. Different types of resolution inconsistency introduced by affine face warping are properly simulated to recreate the artifacts in the DeepFake output pipeline. The smoothed mask is affine warped back to the same sizes as the original faces. Training variety is increased by modifying the many aspects of image appearance: light, contrast, blur, and sharpness.

Two DeepFake video datasets are used to test the models (UADFV and DeepFakeTIMIT). UADFV dataset contains 98 videos in which 50% videos are real and 50% are fake. Each video having a single subject that lasts about 11 s. The models are evaluated on UADFV dataset using Area Under Curve (AUC) metric. ResNet networks outperform VGG16 by around 10%. Among the ResNet networks, ResNet50 has the best results. The models are also evaluated on DeepFakeTIMIT dataset. This dataset includes two sets of fake videos, each of which is generated with a lower quality (LQ) input/output matrix of 64 × 64 pixels and higher quality (HQ) with 128 × 128 pixels. Each fake video collection contains 32 subjects, each subject has 10 videos of swapped and lasts 4 s.

5.6. Two-stream neural network

Zhou et al. [49] propose a two-stream network to detect altered faces. GoogLeNet is trained to detect tampered artefacts as part of a face classification stream, and a patch-based triplet network is trained as part of a second stream to take use of features collecting local noise residuals and camera characteristics. A collection of 2010 altered face images is created using two distinct face swapping programs. Fig. 11 shows the architecture of two-stream neural network.

The GoogLeNet-based face classification stream, which serves as a tampered face classifier, is trained on both tampered

Table 9
AUC values of models.

Models	AUC (%)
Face Classification network	0.854
Patch Triplet network	0.875
Two-stream neural network	**0.927**

and genuine face images. The patch triplet stream, which is based on patch level steganalysis features, captures low-level camera properties and local noise residuals. Steganalysis features are extracted, and the patch triplet network is trained to refine them, instead of directly using steganalysis features. Integrating both the streams yields excellent face tamper detection results by exposing high-level tampering artifacts and low-level noise residuals features.

5.6.1. Face classification stream

Face manipulation causes artifacts, so (such as sharp edges on the mouth or distorted regions on the forehead), the visual details contained in the tampered face is crucial for tampered face detection. A deep CNN model is trained as a classifier to classify whether a face is tampered or not.

5.6.2. Patch triplet stream

The patch triplet stream is used to retrieve insightful clues hidden in in-camera processing for accurate altered face recognition. Co-occurrence-based local features (such as steganalysis features) capture this concealed information and are used to identify image splicing.

Face Classification stream 229 × 229 face images are passed as input and GoogLeNet inception V3 model is used as a classifier. After 8 epochs, the learning rate α is reduced by a factor of two. The batch size is fixed to 32, and all CNN layers are pre-trained using ImageNet weights before the training procedure is finished after 16000 steps. In Patch Triplet Stream, every triplet consists of three 128 × 128 patches, and each patch with 5514D steganalysis features. The triplet network is split into two layers, with 1024 neurons in the first layer and 512 neurons in the second. The network's output is then L2 normalised. The initial learning rate α is set to 0.1 during training and subsequently decreased by 2 per 8 epochs.

5.6.3. Dataset and results

FaceSwap training subset is used to train the model, while SwapMe test subset is used to test it. Because FaceSwap's tampering quality is not as good as of SwapMe dataset. 1400 real images and 705 tampered images are used for training, while 300 real images and 300 tampered images are used for testing. The two-stream network is evaluated on FaceSwap and SwapMe datasets using Area Under Curve (AUC) metric. The AUC performance of two-stream network in comparison with other models is given in Table 9.

5.7. Recurrent convolution strategies (RCS)

RCS [50] detect tampered faces in videos using recurrent convolutional models. Taking use of the stream's temporal details, it is possible to identify tampered faces in videos. Recurrent convolutional networks are a sort of DL technique that has been proven to extract temporal information from image streams. Cropping and aligning faces from video frames is the basic technique for manipulation detection, accompanied by manipulation identification over the pre-processed facial region.

5.7.1. Face preprocessing

Masks provided by [43] are used to crop face regions. Face alignment is employed, as alignment of faces is very helpful in face recognition. Explicit face alignment utilising facial landmarks and implicit face alignment using the Spatial Transformer Network are also employed. The idea is to pass tightly aligned copped faces across video frames as input to the recurrent convolutional model. A four-degree-of-freedom resemblance transformation that accounts for isotropic dimension, in-plane rotation, and 2D translation is used to align face images.

5.7.2. Spatial Transformer Network (STN)

An STN uses learnable affine transformation parameters to perform spatial alignment of data. In DL networks, it is inserted between feature maps. A localisation net, a grid generator, and a sampler are the three components of STN. The grid generator and sampler warp the input feature map with the affine parameters to generate the output feature maps, while the localisation net predicts the affine transformation parameters.

5.7.3. Manipulation detection

Recurrent convolutional network detects manipulation, where sequence of frames from a video is passed as input to the network. The idea behind this model is to take advantage of temporal differences between frames. Since image manipulations are done frame by frame, temporal inconsistencies are anticipated. For the CNN components of the model, ResNet and DenseNet are used. FaceForensics++ is a low-resource data collection with 1,000 videos, and the authors utilise pre-trained XceptionNet with preset feature extraction layers to avoid overfitting. ResNet is chosen for end-to-end trainability because it is simple to learn. Manipulation artifacts also include low-level characteristics (such as broken jawlines and twisted pupils) that do not need high-level face semantic elements. DenseNet is an excellent CNN architecture since it extracts features at several hierarchical levels. Fig. 12 illustrates the pipeline of the recurrent convolutional network model.

5.7.4. Dataset and results

Adam optimiser with 1e-4 learning rate are used as hyper-parameters during the model training. The model is trained on FaceForensics++ dataset, and is evaluated on DeepFake, F2F and FaceSwap datasets as shown in Table 10.

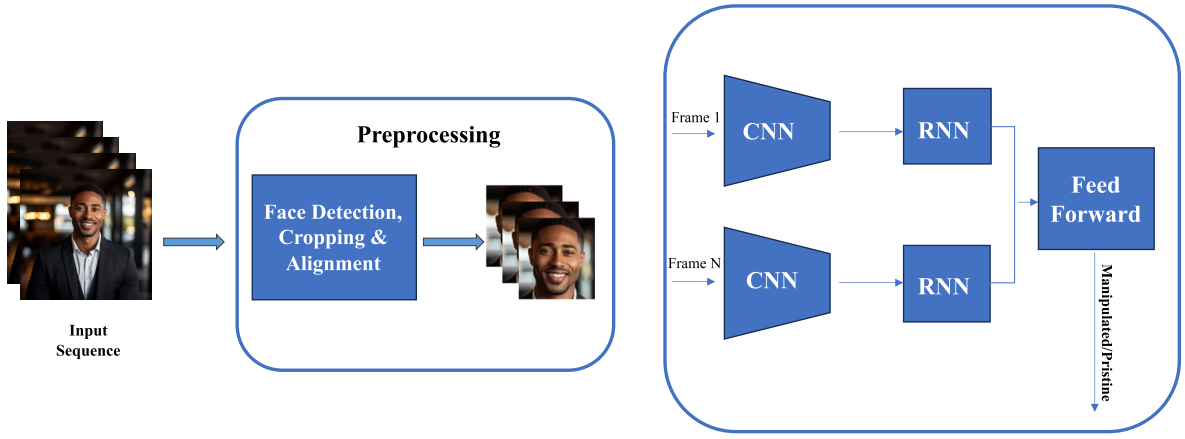


Fig. 12. Pipeline of the recurrent convolutional network model.

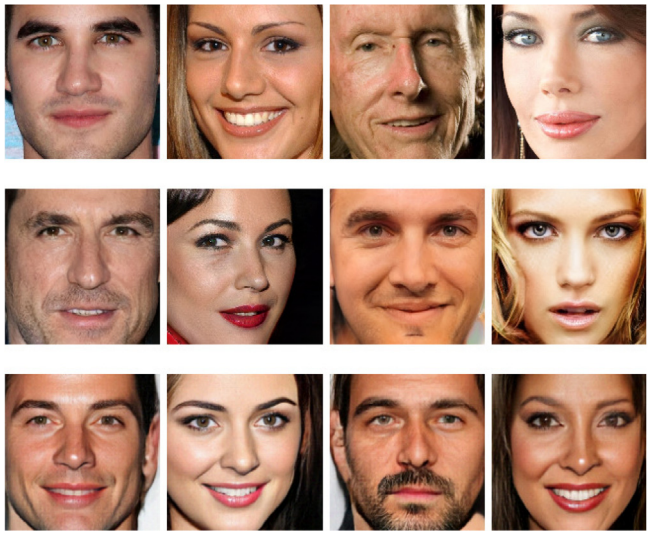


Fig. 13. Examples of dataset used: CelebA [51], ProGAN [52], Glow [53].

Table 10
Results of the recurrent convolutional architecture.

Dataset	Accuracy (%)
DF	96.9
F2F	94.35
FaceSwap	96.3

5.8. Exploiting visual artifacts

Matern et al. [54] proposed a technique to detect DeepFake and face manipulations using simple visual artifacts. To detect visual artifacts, the work is focused on eyes, teeth and facial contours. These visual characteristics are used as an easy, practical bridge technology to identify manipulations. We illustrate the detailed results for DeepFake detection using various methods in Table 11.

5.8.1. Dataset of generated faces

A dataset of created faces, captured DeepFakes, and photographs demonstrating F2F manipulations are used in the evaluation, as shown in Fig. 13. A dataset is collected to classify generated faces. To extract the faces of actual people. The CelebA dataset [51] is used to build face crops. Face crops are extracted with a height of at least 400 pixels to produce samples that qualitatively match the effects of the generation methods. Faces that are categorised as frontal and have a high level of trust are considered only. Face crops are extracted from the ProGAN [52] given samples for generated faces. The process generates images with a resolution of 1024 × 1024 pixels. The images are cropped to the face area using a face detector, which removes the background. The image height after cropping ranges between 446 and 642 pixels. As training and development data, 2000 samples are extracted for the real and generated classes. 1000 random faces are generated using Glow [53] as additional test data and to validate generalisation to a different method. The created images have a 256 × 256-pixel resolution. The height after cropping ranges between 124 and 180 pixels.

5.8.2. Detection pipeline

DF, F2F pictures, and a collection of basic characteristics are used for identifying produced faces. Differences in eye colour are utilised to identify produced faces. Common computer vision methods are used to extract colour attributes from each eye. Each input image is analysed for facial landmarks. To guarantee that all samples are processed at the same resolution, the images are cropped to the face area and scaled to a height of 768 pixels after detection. To calculate eye colour attributes, iris pixels are detected. To identify the Iris displayed as a circle inside the eye's landmarks, Hough Circle Transformation and Canny edge detection are used.

For DeepFake detection, missed reflections and information in the eye and teeth regions are exploited, as shown in Fig. 14. Face landmarks are detected, and the input image is cropped to the facial region. To allow for sizes ranging from 200 pixels to 768 pixels, all data is resized to 256 pixels in height. In the

Table 11
Performance of DeepFake Detection Methods.

Study	Features	Classification methods	Databases	Best performance
Convolutional LSTM	Image Temporal Information	CNN + RNN	own	Acc. = 97.1%
XceptionNet	Image-related Steganalysis	CNN	FF++ (DeepFake, LQ) FF++ (DeepFake, HQ) FF++ (DeepFake, RAW) Celeb-DF	Acc. ≈ 94.0% Acc. ≈ 98.0% Acc. ≈ 100.0% AUC = 65.5%
MesoNet	Mesoscopic Level	CNN	FF++ (DeepFake, LQ) FF++ (DeepFake, HQ) FF++ (DeepFake, RAW) Celeb-DF	Acc. ≈ 90.0% Acc. ≈ 94.0% Acc. ≈ 98.0% AUC = 54.8%
Capsule	Image-related	Capsule Network	DeepFakeTIMIT (LQ) DeepFakeTIMIT (HQ) FF++ (DeepFake) Celeb-DF	AUC = 78.4% AUC = 74.4% Acc. = 92.17% AUC = 57.5%
Face Warping Artifacts	Face Warping Artifacts	CNN	UADFV DeepFakeTIMIT (LQ) DeepFakeTIMIT (HQ)	AUC = 97.4% AUC = 99.9% AUC = 93.2%
Two-Stream	Image related Steganalysis	CNN, SVM	UADFV DeepFakeTIMIT (LQ) DeepFakeTIMIT (HQ)	AUC = 85.1% AUC = 83.5% AUC = 73.5%
Recurrent Convolutional	Image + Temporal Information	CNN + RNN	FF++ (DeepFake, LQ)	AUC = 96.9%
Visual Artifacts	Visual Artifacts	Logistic Regression, MLP	own	AUC. = 85.1%

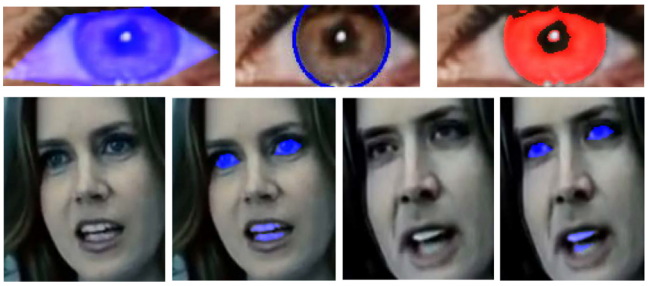


Fig. 14. Pipeline to process face crops and segmentations.

ocular area, the land markings (convex hull of the appropriate landmarks) are used to segment the visual field. The teeth are segmented using gray-scale pictures. K-Means clustering groups the pixels in the mouth landmark's convex hull based on their brightness. Every pixel in the brilliant cluster has teeth. A sample is rejected if fewer than 1% of the mouth pixels are labelled as teeth. To produce feature vectors for each sample, all pixels in the eyes, teeth, and the entire picture are averaged. These descriptors are classified using two distinct models. The logistic regression model is provided as an example of a "off-the-shelf" classifier. A small neural network is developed using a short training time and hyperparameter tuning as a higher capacity model. With three layers, each with 64 nodes and ReLU activation functions, the neural network is fully connected. With a learning rate α of 0.1, the model is trained with an ADAM optimiser and regularised with an L2 penalty. The classifiers are trained using the eye feature vector, the teeth feature vector, a 16-dimensional feature vector including the eye and tooth characteristics, and the feature vector obtained from the full-face crop.

Table 12
AUC values for the classification of the DeepFake using Visual Artifacts.

Classifier model	Eyes acc (%)	Teeth acc (%)	Eyes & teeth acc (%)
MLP-AUC	0.820	0.625	**0.851**
LogReg-AUC	0.775	0.625	0.784

5.8.3. *Classification results*

Table 12 discusses the performance of DeepFake detection methods on all three types of manipulations. The proposed models are evaluated on test data. LogReg gives AUC values of 0.775 (eyes), 0.625 (teeth) and 0.784 (eyes and teeth).

5.8.4. *XceptionNet*

XceptionNet, a CNN based architecture achieved highest accuracy. CNNs are best for feature extraction and identification. Most of the DeepFake detection methods use CNN for feature extraction, but XceptionNet is better than other methods because of its architecture, hyperparameters and the database used for training and validation. The 36 convolutional

layers that make up the Xception architecture are used to extract features. The 36 convolutional layers are grouped into 14 modules, which all have linear residual connections, except for the first and last modules. XceptionNet hyperparameters are set at a learning rate α of 0.0002 and a batch-size of 32. The model is trained using the Adam optimiser. The dataset used for training and validation of the model is FaceForensics++ dataset which is one of the biggest datasets available for DeepFake detection. XceptionNet achieves the highest accuracy of 98% on FaceForensics++ dataset. We provide a set of hyperparameters that must be kept in mind while training a DL-based model to detect DeepFake. These hyperparameters include (i) Minimum number of hidden layers in the network should be 2–4 in a model (ii) Number of hidden units (neurons) for each hidden layer should be in the range of 50 to 100. (iii) Learning rate $\alpha = 0.0001$ - 1 (iv) Hyperparameter for exponentially weighted averages $\beta = 0.9$ - 0.999.

These parameters help researchers develop more robust and efficient models in the domain of computer vision and DL. The technology behind DeepFake creation is getting better and DeepFake detection methods need to be improved for better detection and classification. The existing datasets are not sufficient for models to be trained and achieve the desired accuracy. To further improve DeepFake detection methods, more quality datasets are required.

6. Conclusion

In conclusion, this paper provides a comprehensive analysis of DeepFake video detection techniques, highlighting their strengths, limitations, architectures, and performances. DL, while being instrumental in addressing complex challenges across various domains, including computer vision and data analytics, also introduces threats to privacy and security through the development of DeepFake software. Recognising the urgency of addressing this issue, researchers have proposed several DL-based methods to detect DeepFake videos. After a detailed analysis of the current literature, we find that XceptionNet and MesoNet outperform in terms of accuracy on F2F, FaceSwap and NeuralTextures datasets, while Face Warping Artifacts perform better in terms of AUC.

Through our analysis, we evaluate the pros and cons of these techniques, shedding light on the evolving landscape of DeepFake detection. Among the existing literature, XceptionNet technique outperforms the rest in DeepFake detection. Moreover, we discussed the architectures of these methods, elucidating their underlying mechanisms and computational frameworks. Finally, we have highlight the importance of hyperparameter tuning in enhancing the overall accuracy and efficiency of DeepFake detection models.

CRediT authorship contribution statement

Rozi Khan: Writing – original draft, Visualization, Formal analysis, Methodology. **Muhammad Sohail:** Resources, Software. **Imran Usman:** Formal analysis, Resources, Visualization. **Moid Sandhu:** Conceptualization, Data curation. **Mohsin Raza:** Validation, Visualization. **Muhammad Azfar Yaqub:** Writing – review & editing, Funding acquisition, Writing – original draft. **Antonio Liotta:** Funding acquisition, Supervision.

Declaration of competing interest

The authors declare that there is no conflict of interest in this paper.

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References

[1] M. Westerlund, The emergence of deepfake technology: A review, Technol. Innov. Manag. Rev. 9 (11) (2019).
[2] M.S. Rana, M.N. Nobi, B. Murali, A.H. Sung, Deepfake detection: A systematic literature review, IEEE Access 10 (2022) 25494–25513.
[3] P. Korshunov, S. Marcel, Deepfakes: A new threat to face recognition? Assessment and detection, 2018, arXiv preprint arXiv:1812.08685.
[4] A. Deshmukh, S.B. Wankhade, Deepfake detection approaches using deep learning: A systematic review, Intell. Comput. Network.: Proc. IC-ICN 2020 (2020) 293–302.
[5] P. Yu, Z. Xia, J. Fei, Y. Lu, A survey on deepfake video detection, Iet Biometrics 10 (6) (2021) 607–624.
[6] M.S. Rana, M.N. Nobi, B. Murali, A.H. Sung, Deepfake detection: A systematic literature review, IEEE Access 10 (2022) 25494–25513.
[7] J.W. Seow, M.K. Lim, R.C. Phan, J.K. Liu, A comprehensive overview of deepfake: Generation, detection, datasets, and opportunities, Neurocomputing 513 (2022) 351–371.
[8] S.R. Ahmed, E. Sonuç, M.R. Ahmed, A.D. Duru, Analysis survey on deepfake detection and recognition with convolutional neural networks, in: 2022 International Congress on Human-Computer Interaction, Optimization and Robotic Applications, HORA, IEEE, 2022, pp. 1–7.
[9] L.A. Passos, D. Jodas, K.A. da Costa, L.A.S. Júnior, D. Rodrigues, J. Del Ser, D. Camacho, J.P. Papa, A review of deep learning-based approaches for deepfake content detection, 2022, arXiv preprint arXiv:2202.06095.
[10] A. Heidari, N. Jafari Navimipour, H. Dag, M. Unal, Deepfake detection using deep learning methods: A systematic and comprehensive review, Wiley Interdiscipl. Rev.: Data Min. Knowl. Discov. (2023) e1520.
[11] A. Malik, M. Kuribayashi, S.M. Abdullahi, A.N. Khan, DeepFake detection for human face images and videos: A survey, Ieee Access 10 (2022) 18757–18775.
[12] X. Liu, Y. Yu, X. Li, Y. Zhao, MCL: Multimodal contrastive learning for deepfake detection, IEEE Trans. Circuits Syst. Video Technol. (2023).
[13] A. Raza, K. Munir, M. Almutairi, A novel deep learning approach for deepfake image detection, Appl. Sci. 12 (19) (2022) 9820.
[14] D. Pan, L. Sun, R. Wang, X. Zhang, R.O. Sinnott, Deepfake detection through deep learning, in: 2020 IEEE/ACM International Conference on Big Data Computing, Applications and Technologies, BDCAT, IEEE, 2020, pp. 134–143.
[15] M.S. Rana, A.H. Sung, Deepfakestack: A deep ensemble-based learning technique for deepfake detection, in: 2020 7th IEEE International Conference on Cyber Security and Cloud Computing (CSCloud)/2020 6th IEEE International Conference on Edge Computing and Scalable Cloud, EdgeCom, IEEE, 2020, pp. 70–75.
[16] A. Singh, A.S. Saimbhi, N. Singh, M. Mittal, DeepFake video detection: A time-distributed approach, SN Comput. Sci. 1 (4) (2020) 212.
[17] D. Güera, E.J. Delp, Deepfake video detection using recurrent neural networks, in: 2018 15th IEEE International Conference on Advanced Video and Signal Based Surveillance, AVSS, IEEE, 2018, pp. 1–6.

[18] A. Mitra, S.P. Mohanty, P. Corcoran, E. Kougianos, A machine learning based approach for deepfake detection in social media through key video frame extraction, SN Comput. Sci. 2 (2021) 1–18.
[19] S. Agarwal, H. Farid, T. El-Gaaly, S.-N. Lim, Detecting deep-fake videos from appearance and behavior, in: 2020 IEEE International Workshop on Information Forensics and Security, WIFS, IEEE, 2020, pp. 1–6.
[20] C. Zhao, C. Wang, G. Hu, H. Chen, C. Liu, J. Tang, ISTVT: Interpretable spatial-temporal video transformer for deepfake detection, IEEE Trans. Inf. Forensics Secur. 18 (2023) 1335–1348.
[21] Y. Huang, F. Juefei-Xu, Q. Guo, Y. Liu, G. Pu, Dodging DeepFake detection via implicit spatial-domain notch filtering, IEEE Trans. Circuits Syst. Video Technol. (2023).
[22] G. Li, X. Zhao, Y. Cao, Forensic symmetry for deepfakes, IEEE Trans. Inf. Forensics Secur. 18 (2023) 1095–1110.
[23] I. Perov, D. Gao, N. Chervoniy, K. Liu, S. Marangonda, C. Umé, M. Dpfks, C.S. Facenheim, L. RP, J. Jiang, et al., DeepFaceLab: Integrated, flexible and extensible face-swapping framework, 2020, arXiv preprint arXiv:2005.05535.
[24] S. Lyu, Deepfake detection: Current challenges and next steps, in: 2020 IEEE International Conference on Multimedia & Expo Workshops, ICMEW, IEEE, 2020, pp. 1–6.
[25] N. Zhuang, C. Yang, Few-shot knowledge transfer for fine-grained cartoon face generation, in: 2021 IEEE International Conference on Multimedia and Expo, ICME, IEEE, 2021, pp. 1–6.
[26] A.A. Maksutov, V.O. Morozov, A.A. Lavrenov, A.S. Smirnov, Methods of deepfake detection based on machine learning, in: 2020 IEEE Conference of Russian Young Researchers in Electrical and Electronic Engineering, EIConRus, IEEE, 2020, pp. 408–411.
[27] A. Hilton, Computer vision for human modelling and analysis, Mach. Vis. Appl. 14 (2003) 206–209.
[28] A. Tewari, M. Elgharib, G. Bharaj, F. Bernard, H.-P. Seidel, P. Pérez, M. Zollhofer, C. Theobalt, Stylerig: Rigging stylegan for 3d control over portrait images, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 6142–6151.
[29] D. Smirnov, M. Gharbi, M. Fisher, V. Guizilini, A. Efros, J.M. Solomon, Marionette: Self-supervised sprite learning, Adv. Neural Inf. Process. Syst. 34 (2021) 5494–5505.
[30] Z. Weiguo, Z. Chenggang, Exposing face-swap images based on deep learning and ELA detection, 2019.
[31] X. Chang, J. Wu, T. Yang, G. Feng, Deepfake face image detection based on improved VGG convolutional neural network, in: 2020 39th Chinese Control Conference, CCC, IEEE, 2020, pp. 7252–7256.
[32] T. Zhang, L. Deng, L. Zhang, X. Dang, Deep learning in face synthesis: A survey on deepfakes, in: 2020 IEEE 3rd International Conference on Computer and Communication Engineering Technology, CCET, IEEE, 2020, pp. 67–70.
[33] J. Ren, M. Chai, O.J. Woodford, K. Olszewski, S. Tulyakov, Flow guided transformable bottleneck networks for motion retargeting, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 10795–10805.
[34] M.P. Aylett, D.A. Braude, C.J. Pidcock, B. Potard, Voice puppetry: Exploring dramatic performance to develop speech synthesis, in: Proc. 10th ISCA Speech Synthesis Workshop, 2019, pp. 117–120.
[35] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, M. Nießner, Faceforensics: A large-scale video dataset for forgery detection in human faces, 2018, arXiv preprint arXiv:1803.09179.
[36] Á.F. Gambín, A. Yazidi, A. Vasilakos, H. Haugerud, Y. Djenouri, Deepfakes: Current and future trends, Artif. Intell. Rev. 57 (3) (2024) 64.
[37] A. Heidari, N. Jafari Navimipour, H. Dag, M. Unal, Deepfake detection using deep learning methods: A systematic and comprehensive review, Wiley Interdiscipl. Rev.: Data Min. Knowl. Discov. 14 (2) (2024) e1520.
[38] A. Gupta, D. Pandey, Deepfake videos generation and detection: A comprehensive survey, in: 2024 IEEE International Conference on Computing, Power and Communication Technologies, Vol. 5, IC2PCT, IEEE, 2024, pp. 1939–1944.
[39] Z. Guo, Z. Jia, L. Wang, D. Wang, G. Yang, N. Kasabov, Constructing new backbone networks via space-frequency interactive convolution for deepfake detection, IEEE Trans. Inf. Forensics Secur. (2023).
[40] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, Z. Wojna, Rethinking the inception architecture for computer vision, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 2818–2826.
[41] I. Laptev, M. Marszalek, C. Schmid, B. Rozenfeld, Learning realistic human actions from movies, in: 2008 IEEE Conference on Computer Vision and Pattern Recognition, IEEE, 2008, pp. 1–8.
[42] K. Da, A method for stochastic optimization, 2014, arXiv preprint arXiv:1412.6980.
[43] A. Rossler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, M. Nießner, Faceforensics++: Learning to detect manipulated facial images, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 1–11.
[44] J. Thies, M. Zollhofer, M. Stamminger, C. Theobalt, M. Nießner, Face2face: Real-time face capture and reenactment of rgb videos, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 2387–2395.
[45] Faceswap github, 2024, https://github.com/deepfakes/faceswap/. (Accessed 17 February 2024).
[46] D. Afchar, V. Nozick, J. Yamagishi, I. Echizen, Mesonet: A compact facial video forgery detection network, in: 2018 IEEE International Workshop on Information Forensics and Security, WIFS, IEEE, 2018, pp. 1–7.
[47] H.H. Nguyen, J. Yamagishi, I. Echizen, Use of a capsule network to detect fake images and videos, 2019, arXiv preprint arXiv:1910.12467.
[48] Y. Li, S. Lyu, Exposing deepfake videos by detecting face warping artifacts, 2018, arXiv preprint arXiv:1811.00656.
[49] P. Zhou, X. Han, V.I. Morariu, L.S. Davis, Two-stream neural networks for tampered face detection, in: 2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops, CVPRW, IEEE, 2017, pp. 1831–1839.
[50] E. Sabir, J. Cheng, A. Jaiswal, W. AbdAlmageed, I. Masi, P. Natarajan, Recurrent convolutional strategies for face manipulation detection in videos, Interfaces (GUI) 3 (1) (2019).
[51] Z. Liu, P. Luo, X. Wang, X. Tang, Deep learning face attributes in the wild, in: Proceedings of the IEEE International Conference on Computer Vision, 2015, pp. 3730–3738.
[52] T. Karras, T. Aila, S. Laine, J. Lehtinen, Progressive growing of gans for improved quality, stability, and variation, 2017, arXiv preprint arXiv:1710.10196.
[53] D.P. Kingma, P. Dhariwal, Glow: Generative flow with invertible 1x1 convolutions, 2018, arXiv preprint arXiv:1807.03039.
[54] F. Matern, C. Riess, M. Stamminger, Exploiting visual artifacts to expose deepfakes and face manipulations, in: 2019 IEEE Winter Applications of Computer Vision Workshops, WACVW, IEEE, 2019, pp. 83–92.